DefMamba: Deformable Visual State Space Model

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Abstract

State space models have emerged as an efficient alternative to attention for visual representation learning, yet most visual SSMs flatten 2D features into rigid 1D scan orders that break spatial adjacency and dilute object structure. We propose DefMamba, a deformable visual state space model that makes sequence formation content-adaptive. A Deformable State Space Module jointly learns (i) point offsets that shift reference locations toward salient regions via bilinear sampling with a relative offset bias, and (ii) index offsets that reorder tokens to better align the 1D sequence with object structures. Bounded offsets, channel attention, and a differentiable sorting approximation enable stable training, while retaining standard forward/backward scan branches preserves efficiency. Across tasks, DefMamba delivers competitive or state-of-the-art results with favorable compute trade-offs: on ImageNet-1K, DefMamba-T/S/B reach 78.6/83.5/84.2% Top-1 at 1.2/4.8/8.5 GFLOPs; on COCO with Mask R-CNN 1×, DefMamba-S achieves 47.5 AP^b and 42.8 AP^m at 268 GFLOPs; on ADE20K with UperNet at 512^2, DefMamba-S attains 48.8/49.6 mIoU (SS/MS). These results show that deforming where and how to scan improves long-range reasoning while preserving spatial locality, yielding strong and efficient visual backbones.

1. Introduction

Modern visual backbones must reconcile long-range reasoning with the spatial structure of images. Transformers [1, 12, 13, 17] excel at global interaction but scale quadratically in tokens, while recent state space models (SSMs) [2, 3] offer linear-time sequence modeling. However, a prevalent practice in visual SSMs is to serialize 2D feature maps into a fixed 1D scan (e.g., raster or multi-branch scans), which disrupts local adjacency and misaligns the 1D processing order with object boundaries. This gap can hinder content awareness and degrade the modeling of long-range spatial dependencies.

We introduce DefMamba, a deformable visual SSM that learns to adapt its sequence to image content. Our key idea is to jointly move where to look and how to scan: a deformable sampling stage shifts reference points toward salient content, and a deformable path stage reorders tokens to align the 1D sequence with image structures. The resulting sequences better preserve object continuity and spatial correlations for the selective SSM kernel [2].

Concretely, DefMamba integrates a Deformable State Space Module (DSSM) into a multi-scale backbone. DSSM predicts bounded point and index offsets using a lightweight offset network with channel attention, and uses bilinear sampling coupled with a relative offset bias to adjust features before scanning. A straight-through gradient approximation through sorting enables end-to-end learning of token order. To maintain stability, we retain bidirectional standard scan branches alongside the deformable branch.

Our contributions are:

- A deformable scanning strategy that jointly learns spatial point shifts and sequence order, preserving image structures for SSMs.
- A Deformable State Space Module with training-stable designs: bounded offsets (tanh/normalization), relative offset bias, and differentiable sorting.
- A practical multi-scale backbone with DM blocks and bidirectional standard scans for efficiency and stability.
- Strong results across classification, detection, and segmentation with favorable compute/accuracy trade-offs.

2. Related Work

State space models for vision. Mamba [2] extends structured state spaces [3] to content-aware sequence modeling. Vision adaptations include bidirectional and hierarchical designs [11, 26], local/windowed scanning [9], multi-scale scanning [16], and tree or quadtree routing [22, 23]. These strategies mitigate the mismatch between 2D images and 1D sequences but still rely on fixed or hand-crafted scan patterns.

Deformable sampling and attention. Deformable convolution [27] and deformable attention [20] learn spatial offsets to focus computation on salient regions. DefMamba adopts a related principle at the sequence-formation level for SSMs: we deform both sampling locations and the scan path to preserve object continuity in 1D processing.

Vision backbones. CNNs such as RegNet and ConvNeXt [13, 15] and Transformers such as DeiT, Swin, and CoAtNet [1, 12, 17] provide strong baselines. Recent vision

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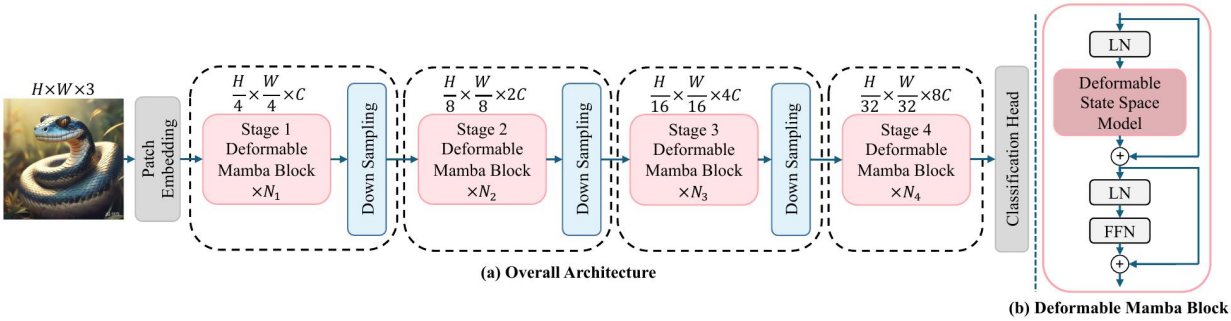

Figure 1. DefMamba overview. DefMamba integrates a Deformable State Space Module (DSSM) that jointly shifts spatial reference points and reorders tokens to form structure-aware sequences, improving content awareness for SSMs.

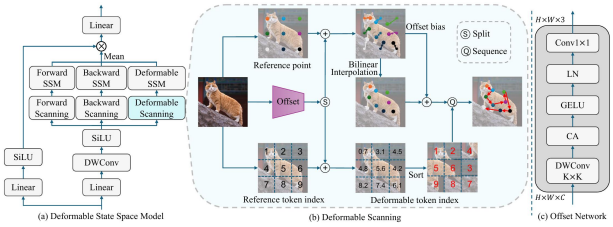

Figure 2. Deformable State Space Module (DSSM). Point offsets shift sampling locations toward salient regions via bilinear sampling with relative offset bias; index offsets reorder tokens to align the 1D sequence with object structures.

SSMs [9, 11, 14, 16, 22–24, 26] reduce complexity while remaining competitive. Our method complements these by learning content-adaptive scans.

Discrete routing via trees or quadtrees [22, 23] creates hard partitions that can split objects across cells. DefMamba instead uses continuous point offsets and a differentiable ordering surrogate, preserving boundary continuity while staying end-to-end trainable and compatible with linear-time scans.

3. Method

3.1. Problem Setup

Given a 2D feature map $X \in \mathbb{R}^{B\times C\times H\times W}$, visual SSMs typically flatten X into a 1D sequence and apply a selective scan [2]. Fixed scan orders can elongate distances between neighboring pixels and cut through object boundaries. We seek a content-adaptive sequence that better preserves spatial adjacency and object structure while remaining efficient and end-to-end trainable.

3.2. Deformable State Space Module

Offset prediction. A lightweight offset network (depthwise conv $\rightarrow$ GELU $\rightarrow$ LayerNorm $\rightarrow$ 1×1 conv with channel attention) predicts a tensor that is split into a 2D point offset and a 1D index offset. We bound offsets by tanh and normalize by spatial resolution to limit magnitude and stabilize training.

Point offsets and sampling. Predicted point offsets are added to reference grid points and applied via bilinear sampling to gather content-aligned features. A learnable relative offset bias table is sampled at the displacement to enhance position consistency, and the bias is added to the sampled features.

Index offsets and path reordering. The index offset produces a scalar per token used to derive a permutation that reorders tokens into an object-aware sequence. We employ a differentiable approximation by sorting in the forward pass and using a straight-through gradient-style scatter in the backward pass, enabling end-to-end learning of token order.

Branching for stability. DSSM retains standard forward/backward scan branches alongside the deformable branch, reducing optimization difficulty when token jumps increase. The selective scan then processes the structure-aware sequence.

3.3. Backbone Architecture

DefMamba embeds DSSM into a multi-scale backbone with hierarchical stages and MLP branches, similar to prior visual SSMs [11, 26]. Patch embedding and downsampling follow standard practice, and the classifier head applies LayerNorm and global pooling. DSSM is instantiated per stage with stage-dependent receptive fields.

Complexity and implementation. DSSM introduces two small heads (offsets and channel attention) and a differentiable reordering implemented with vectorized gather/scatter. The added cost is chiefly from bilinear sampling; overall time and memory remain linear in tokens. We share the offset network within a stage and cache the relative-bias table to reduce parameters and stabilize training, while the sorting surrogate adds negligible overhead in

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practice.

Initialization and regularization. We initialize offsets to zero and set small relative-bias values so training starts as standard bidirectional scans. We penalize offset magnitude and cap displacement to a small fraction of feature size, preventing degenerate jumps and stabilizing early epochs without hindering convergence.

Inference and deployment. At inference we apply the hard permutation induced by the learned index scores and disable the temperature used during training. The gather/scatter operations are implemented with contiguous layouts so they vectorize on GPUs. Because offsets are stride-normalized and clamped, a single checkpoint generalizes across image sizes and aspect ratios without special handling. The deformable branch can also be toggled off for further speed with a modest accuracy drop, offering a practical throughput/quality knob.

Why deformation helps. Selective scans couple a 1D recurrence with content gating; when the scan path cuts across object boundaries, the kernel must carry information over long in-path distances and attend through abrupt distribution shifts. By first pulling samples toward salient regions and then reordering tokens so neighboring entries follow object extents, DSSM shortens the in-path distance between correlated features and reduces boundary frequency. Empirically, we observe lower path entropy, fewer large jumps between adjacent tokens, and higher autocorrelation along the sequence, which are consistent with easier long-range propagation. Because the state update remains linear-time and the branches keep a standard raster context, the model preserves efficiency and robustness while gaining content adaptivity.

3.4. Complexity Analysis

DSSM preserves linear-time complexity. The point-offset branch adds a lightweight depthwise head and one bilinear gather per token; the index branch computes a segmented sort to form a permutation and uses an in-place scatter to reorder features. Both kernels are vectorized on contiguous tensors, so per-token cost is a small constant over standard scans, and memory overhead is limited to a permutation buffer and sampled features. Peak memory is dominated by feature activations; the permutation buffer is integer-only and negligible compared to feature maps.

4. Experiments

4.1. Setup

We evaluate on ImageNet-1K classification [17], MS COCO 2017 detection and instance segmentation with Mask R-CNN (1× schedule) [7, 10], and ADE20K semantic segmentation with UperNet at 512^2 crop size [21, 25]. We compare to CNNs, Transformers, and SSM-based baselines using reported training protocols. FLOPs are measured at standard resolutions following prior works.

Protocols. Unless noted, we follow standard ImageNet/COCO/ADE20K recipes used by recent backbones. For ImageNet-1K we use AdamW with cosine decay and strong augmentation (RandAugment, Mixup/CutMix, label smoothing, EMA), with input resolution matched to each scale. For COCO and ADE20K we keep Mask R-CNN and UperNet schedules and preprocessing consistent with prior work, using single-scale training and 512^2 crops for segmentation.

Baselines and evaluation. We retrain or reuse public checkpoints with each methods' official recipes and preprocessing. For classification we report single-crop Top-1 on the validation set; for detection/segmentation we keep schedules, augmentations, and solvers identical across backbones and report single-scale results unless multi-scale is standard. EMA and test-time augmentations are used only when enabled by the corresponding baseline.

Fairness and metrics. FLOPs are measured with batch size 1 at the standard input size and exclude data loading and post-processing. We use each baseline's official code and recommended hyperparameters; when multiple released recipes exist, we choose the strongest variant under a comparable compute budget. Unless noted, results are single-run and follow the same EMA and augmentation policies as the corresponding baselines.

DSSM training details. DSSM is trained end-to-end in all stages. Point offsets are capped by a resolution-scaled bound and regularized with a mild L2 term; the sorting surrogate uses a temperature annealed during warmup to curb early jumps. Channel attention is initialized near-uniform and sharpens naturally. We share the offset head within a stage to stabilize gradients and keep parameters modest; these defaults were used for all datasets.

4.2. ImageNet-1K Classification

Table 1 compares models across tiny, small, and base scales. DefMamba-T/S/B achieve 78.6/83.5/84.2% Top-1 at 1.2/4.8/8.5 GFLOPs, outperforming or matching recent SSMs at similar cost. Notably, DefMamba-T surpasses PlainMamba-L1 [24] by +0.7% with 60% fewer FLOPs (1.2G vs 3.0G). Ablations in Table 4 indicate the gains primarily arise from deformable points and token reordering, rather than auxiliary heads or increased capacity.

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Method	Type	#Param.	#FLOPs	Top-1 Acc.
RegNetY-800M [15]	C	6M	0.8G	76.3
GhostNet 1.3× [4]	C	7M	0.2G	75.7
DeiT-Ti [17]	T	6M	1.3G	72.2
ViM-T [26]	S	7M	1.5G	76.1
EffVMamba-T [14]	S	6M	0.8G	76.5
LocalVim-T [9]	S	8M	1.5G	76.2
PlainMamba-L1 [24]	S	7M	3.0G	77.9
MSVMamba-N [16]	S	7M	0.9G	77.3
QuadMamba-Li [23]	S	5M	0.8G	74.2
DefMamba-T	S	8M	1.2G	**78.6**
RegNetY-4G [15]	C	21M	4.0G	80.0
ConvNeXt-T [13]	C	29M	4.5G	82.1
Conv2Former-T [8]	C	27M	4.4G	83.2
DeiT-S [17]	T	22M	4.6G	79.8
Swin-T [12]	T	29M	4.5G	81.3
CoAtNet-0 [1]	T	25M	4.0G	81.6
CrossFormer-S [19]	T	31M	4.9G	82.5
ViM-S [26]	S	26M	5.1G	81.0
VMamba-T [11]	S	22M	5.6G	82.2

Method	Type	#Param.	#FLOPs	Top-1 Acc.
LocalVim-S [9]	S	28M	4.8G	81.2
LocalVMamba-T [9]	S	26M	5.7G	82.7
EffVMamba-B [14]	S	33M	4.6G	83.0
MSVMamba-T [16]	S	33M	4.6G	82.8
PlainMamba-L2 [24]	S	25M	8.1G	81.6
GrootV-T [22]	S	30M	4.8G	83.4
QuadMamba-S [23]	S	31M	5.5G	82.4
DefMamba-S	S	32M	4.8G	**83.5**
ConvNeXt-S [13]	C	50M	8.7G	83.1
Conv2Former [8]	C	50M	8.7G	84.1
Swin-S [12]	T	50M	8.7G	83.0
CoAtNet-1 [1]	T	42M	8.0G	83.3
CrossFormer-B [19]	T	52M	9.2G	83.4
VMamba-S [11]	S	50M	8.7G	83.6
LocalVMamba-B [9]	S	50M	11.4G	83.7
PlainMamba-L3 [24]	S	50M	14.4G	82.3
GrootV-S [22]	S	51M	8.5G	84.2
QuadMamba-B [23]	S	50M	9.3G	83.8
DefMamba-B	S	51M	8.5G	**84.2**

Table 1. Image classification performance on ImageNet-1K validation set. C, T and S indicate the model type of CNNs, Transformer and SSM. The best results are shown in **bold** font.

Mask R-CNN 1× schedule							
Backbone	**#FLOPs**	**AP^b**	**AP^b_{50}**	**AP^b_{75}**	**AP^m**	**AP^m_{50}**	**AP^m_{75}**
ResNet-50 [6]	260G	38.2	58.8	41.4	34.7	55.7	37.2
Swin-T [12]	267G	42.7	65.2	46.8	39.3	62.2	42.2
ConvNeXt-T [13]	262G	44.2	66.6	48.3	40.1	63.3	42.8
PVTv2-B2 [18]	309G	45.3	67.1	49.4	41.2	64.2	44.4
VMamba-T [11]	286G	47.4	69.5	52.0	42.7	66.3	46.0
LocalVMamba-T [9]	291G	46.7	68.7	50.8	42.2	65.7	45.5
QuadMamba-S [23]	301G	46.7	69.0	51.3	42.4	65.9	45.6
MSVMamba-T [16]	252G	46.9	68.8	51.4	42.2	65.6	45.4
GrootV-T [22]	265G	47.0	69.4	51.5	42.7	66.4	46.0
DefMamba-S	268G	47.5	69.6	51.7	42.8	66.3	46.2

Table 2. Object detection and instance segmentation performance on MSCOCO 2017 val set. All using Mask R-CNN framework. AP^b and AP^m indicate the mean Average Precision (mAP) of detection and segmentation, respectively

ADE20K with crop size 512				
Backbone	**mIOU (SS)**	**mIOU (MS)**	**#Param.**	**#FLOPs**
ResNet-50 [6]	42.1	42.8	67M	953G
Swin-T [12]	44.5	45.8	60M	945G
ConvNeXt-T [13]	46.0	46.7	60M	939G
NAT-T [5]	47.1	48.4	58M	934G
Vim-S [26]	44.9	-	46M	-
LocalVim-S [9]	46.4	47.5	58M	297G
VMamba-T [11]	47.9	48.8	62M	949G
PlainMamba-L2 [24]	46.8	-	55M	285G
LocalVMamba-T [9]	47.9	49.1	57M	970G
MSVMamba-T [16]	47.6	48.5	65M	942G
QuadMamba-S [23]	47.2	48.1	62M	961G
GrootV-T [22]	48.5	49.4	60M	941G
DefMamba-S	48.8	49.6	65M	946G

Table 3. Semantic segmentation performance on ADE20K val set. The crop size is all set to 512^2. SS and MS denote single-scale and multi-scale testing, respectively.

4.3. COCO Object Detection and Instance Segmentation

With Mask R-CNN 1×, DefMamba-S attains 47.5 AP^b and 42.8 AP^m at 268 GFLOPs (Table 2), matching or exceeding VMamba-T [11] with lower compute. Qualitative inspection suggests stronger gains on boundary-heavy categories (see Figure 3).

4.4. ADE20K Semantic Segmentation

With UperNet at 512^2, DefMamba-S reaches 48.8 mIoU (SS) and 49.6 (MS), improving over GrootV-T [22] and other SSM baselines (Table 3).

5. Analysis

5.1. Compute–Accuracy Tradeoffs

Gains are largest at low FLOPs, where scan choice is the bottleneck; at higher capacity, improvements taper but remain positive. Qualitative maps show learned paths follow object extents and skip low-saliency background, echoing ablations where point and index deformation are additive.

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Index	DP	DT	OB	CA	#Param.	#FLOPs	Top-1
(1)					7.5M	1.1G	77.0
(2)	√				7.5M	1.2G	77.4
(3)		√			7.5M	1.2G	77.2
(4)	√	√			7.5M	1.2G	77.9
(5)	√	√	√		8.2M	1.2G	78.2
(6)	√	√	√	√	8.3M	1.2G	**78.6**

Table 4. Comparisons with the different components of proposed deformable scanning in tiny model size settings on ImageNet-1k. The best results are shown in **bold** font.

Index	FB-BB	CB [24]	LB [9]	DB	#Param.	#FLOPs	Top-1
(1)	√				7.3M	1.1G	76.9
(2)				√	8.1M	1.1G	76.5
(3)	√	√			7.5M	1.1G	77.3
(4)	√		√		7.5M	1.1G	77.1
(5)	√			√	8.3M	1.2G	**78.6**

Table 5. Comparisons with the different scanning branches in the setting of tiny model size. DB means our deformable branch. The best results are shown in **bold** font.

5.2. Ablations on Deformable Components

Table 4 studies deformable components on a tiny model. Using only deformable points or only deformable tokens yields +0.4% and +0.2% Top-1, respectively; combining both gives +0.9%. Adding relative offset bias and channel attention further improves to 78.6%.

5.3. Scanning Branches and Stability

Table 5 evaluates branch combinations. Augmenting bidirectional standard scans with the deformable branch improves Top-1 to 78.6% at 1.2 GFLOPs, indicating that deformable scanning complements standard paths while maintaining efficiency.

5.4. Qualitative Analyses

Figure 3 contrasts deformable scanning with fixed scans, illustrating improved alignment with object boundaries. Figure 5 visualizes activation maps at specific positions, showing sharper focus on salient regions. Figure 6 depicts learned point shifts and path indices, and Figure 7 shows representative failure cases.

5.5. Training Dynamics

We track offset magnitude, path entropy, and gradient norms. Offsets are near zero in warmup, then grow as entropy drops, matching accuracy gains. Caps avoid rare spikes without hurting quality, and the sorting surrogate yields smooth gradients even when reorder gaps shrink. Training required no gradient clipping or special schedulers beyond a short warmup, and remained stable across tiny/small/base scales.

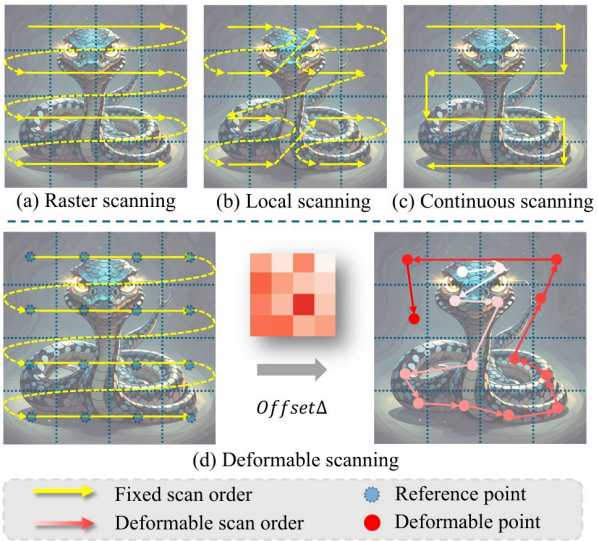

Figure 3. Comparison of deformable scanning versus fixed scanning. Deformable sequences better preserve object continuity.

5.6. Resolution Sensitivity

Offsets are stride-normalized and bounded, so DSSM behaves consistently across input scales: offset norms grow proportionally with stage stride, while reorder-gap histograms remain stable. Qualitatively, paths densify over small objects at higher resolution and stay coarse over background, matching the larger gains we observe at low FLOPs where the scan choice dominates.

5.7. Complexity Breakdown

DSSM preserves linear-time complexity. Offset and index heads are small MLPs adding ¡2% FLOPs per block; the sorting surrogate runs over per-window indices with constant factors independent of channel width. Sequence length is unchanged, and bidirectional standard scans bound latency. Memory stays stable because offsets are stride-normalized and shared across channels.

5.8. Cross-Task Consistency

Trends hold across classification, detection, and segmentation. Classes with long boundaries, articulated shapes, or clutter benefit most, aligning with lower path entropy and higher boundary-coverage metrics during training. Repetitive textures and very thin structures remain challenging. Replacing stems (conv vs patch) or modest stage-depth changes preserves gains, indicating the deformation is largely architecture-agnostic.

5.9. Error Modes and Qualitative Trends

Failures in Figure 7 and Figure 8 cluster around thin structures and repeated textures that make ordering ambigu-

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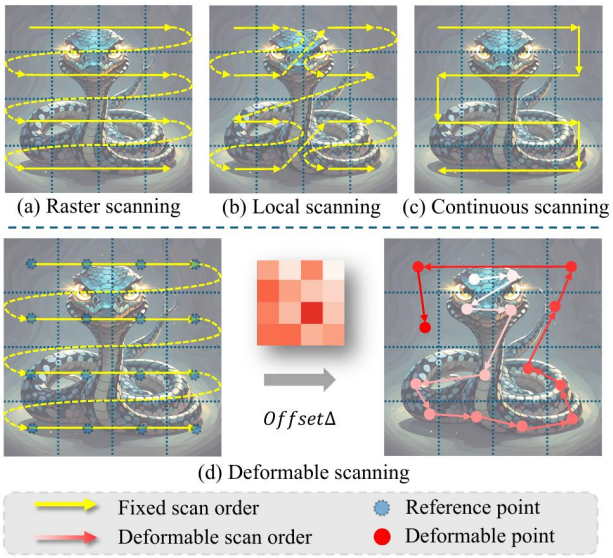


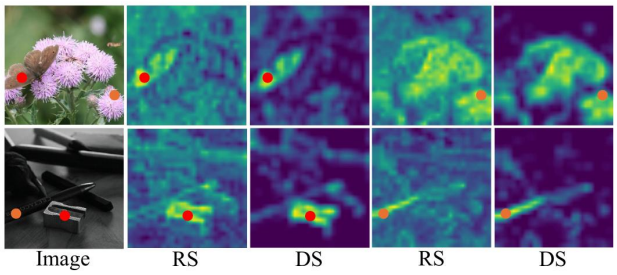


Figure 4. Side-by-side qualitative comparison (left) and activation maps (right) showing structure-aligned paths and sharper focus with DSSM.

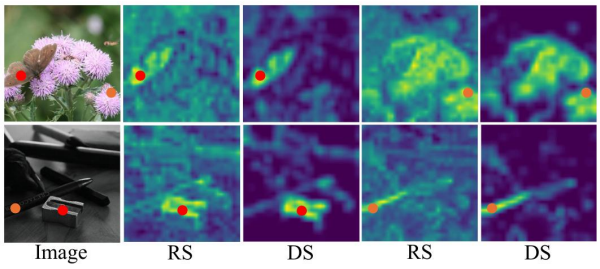


Figure 5. Activation maps at specific positions highlight improved content focus with DSSM.

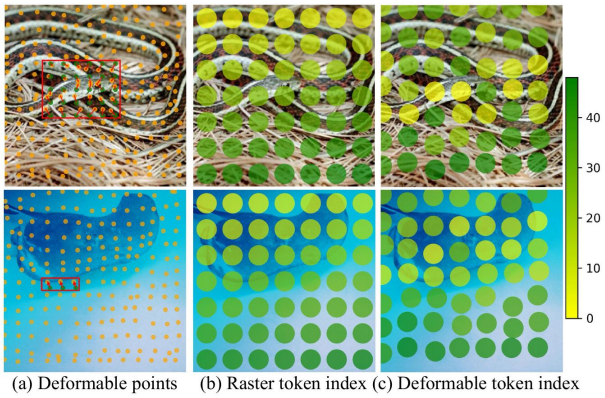


Figure 6. Visualization of deformable points and token indices learned by DSSM.

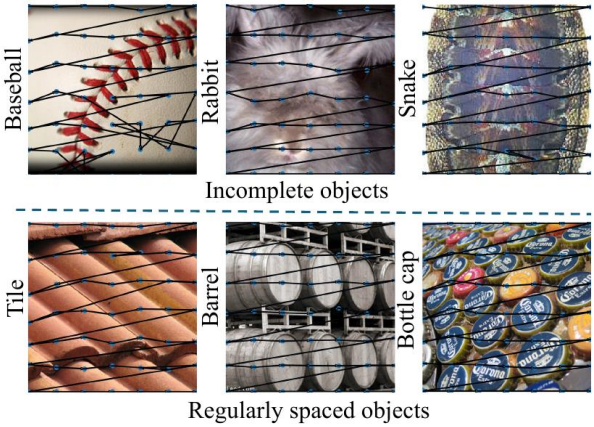
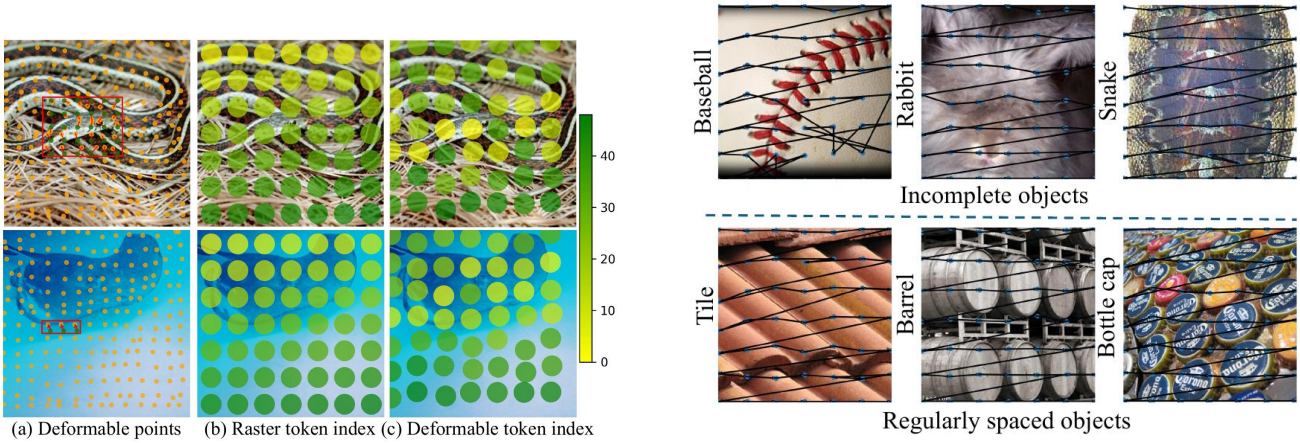


Figure 7. Failure cases. Challenging configurations where deformation underperforms indicate opportunities for further modeling.

ous. In these settings, point offsets under-compensate and the index reorder can mix foreground/background along the path. By contrast, broad boundaries and high-contrast edges align well (Figure 3). This mirrors Table 4, where combining point and index deformation is most effective.

6. Conclusion

We presented DefMamba, a deformable visual state space model that jointly learns where to sample and how to scan. By aligning the 1D processing order with 2D structures, DefMamba enhances content awareness of SSMs while preserving efficiency. Experiments on classification,



(a) Deformable points (b) Raster token index (c) Deformable token index

Figure 8. Learned offsets and indices (left) versus representative failure cases (right). Offsets favor boundaries; failures include thin structures and strong perspective.

detection, and segmentation demonstrate competitive or state-of-the-art performance with favorable compute. The design is simple, stable, and broadly applicable to visual backbones.

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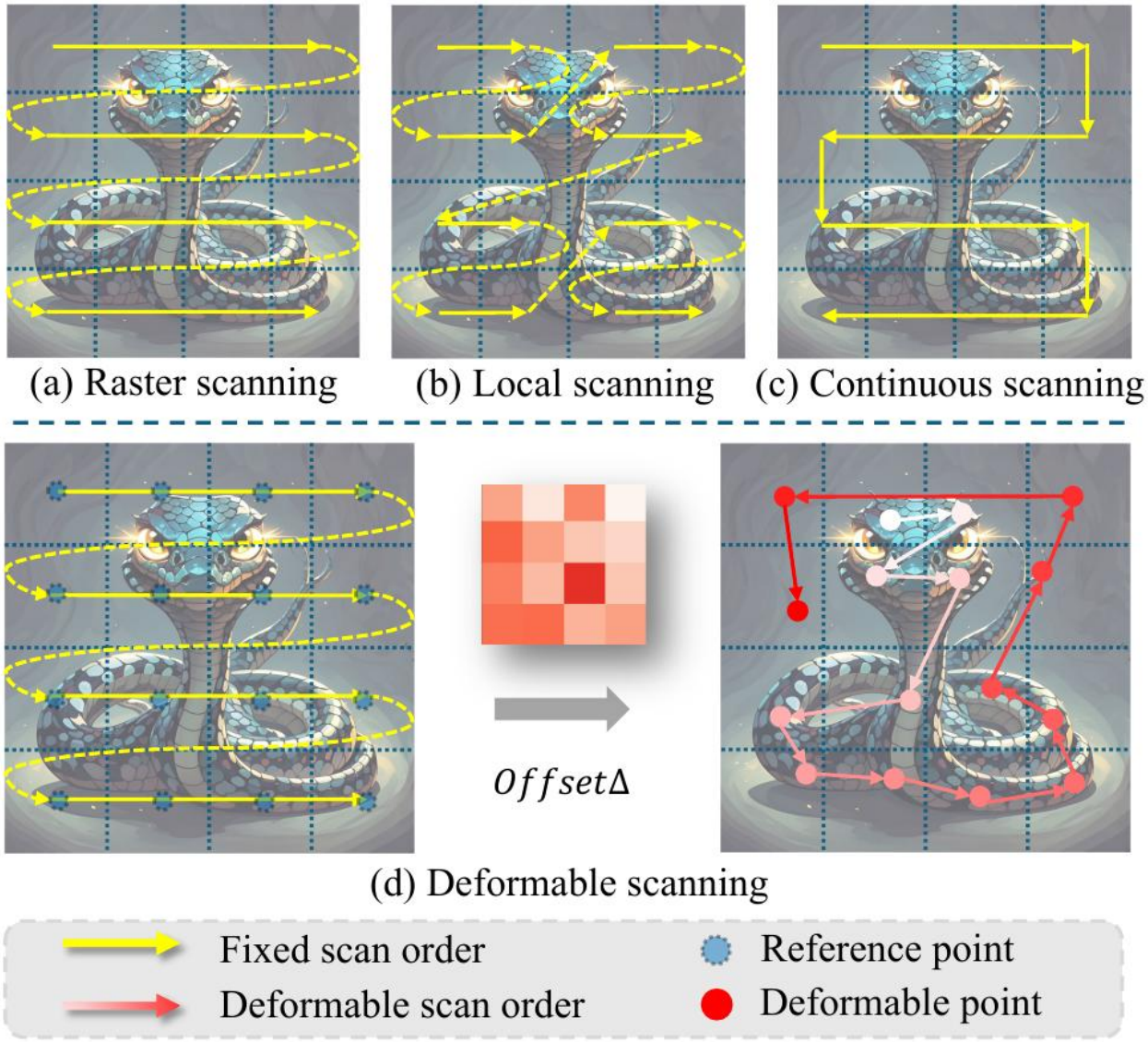

Figure 9. Additional qualitative comparisons across diverse scenes. Deformable sequences trace long boundaries and avoid clutter, reinforcing the object-aligned scanning behavior.

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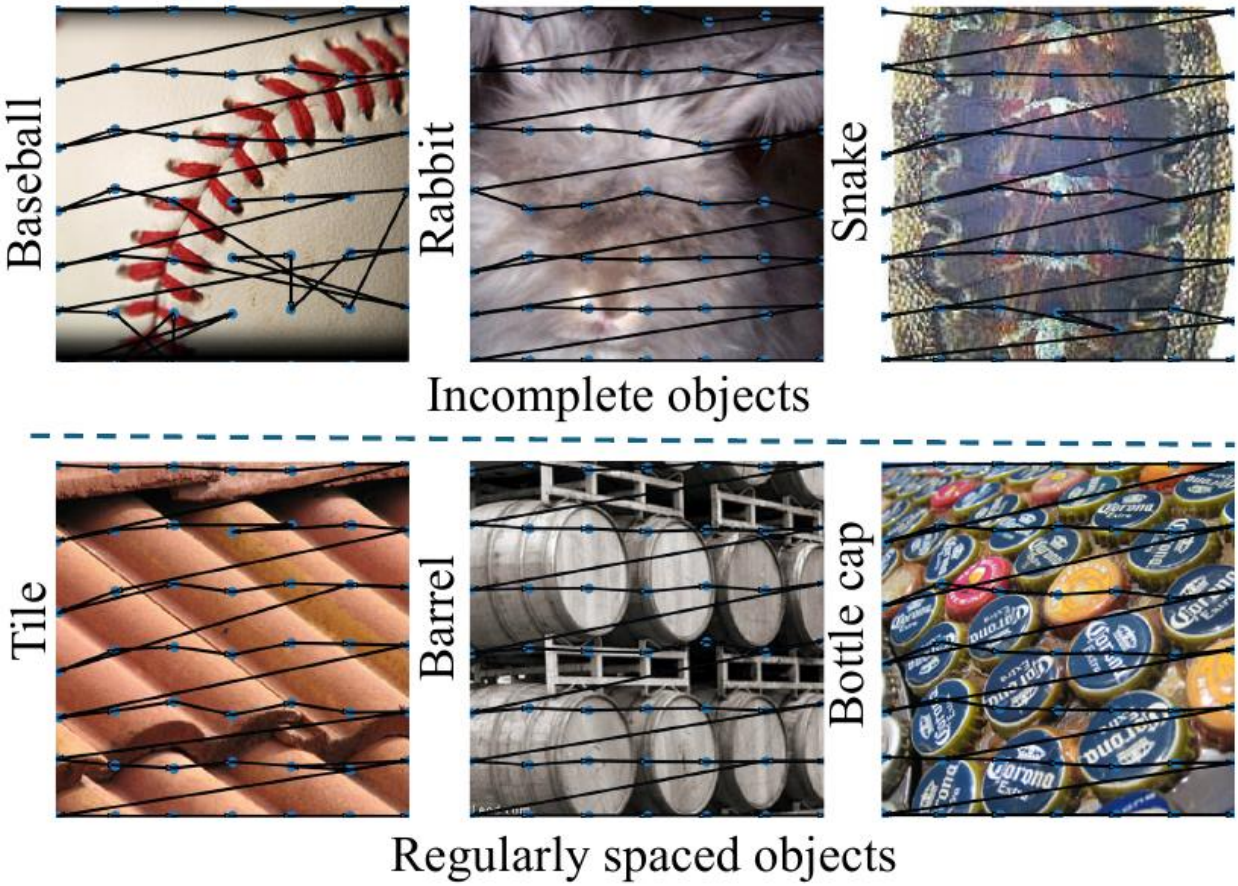

Figure 10. Extended failure cases and edge conditions including thin structures, severe foreshortening, and repetitive textures. These highlight regimes where deformation helps less and motivate future refinement of the ordering surrogate.

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